

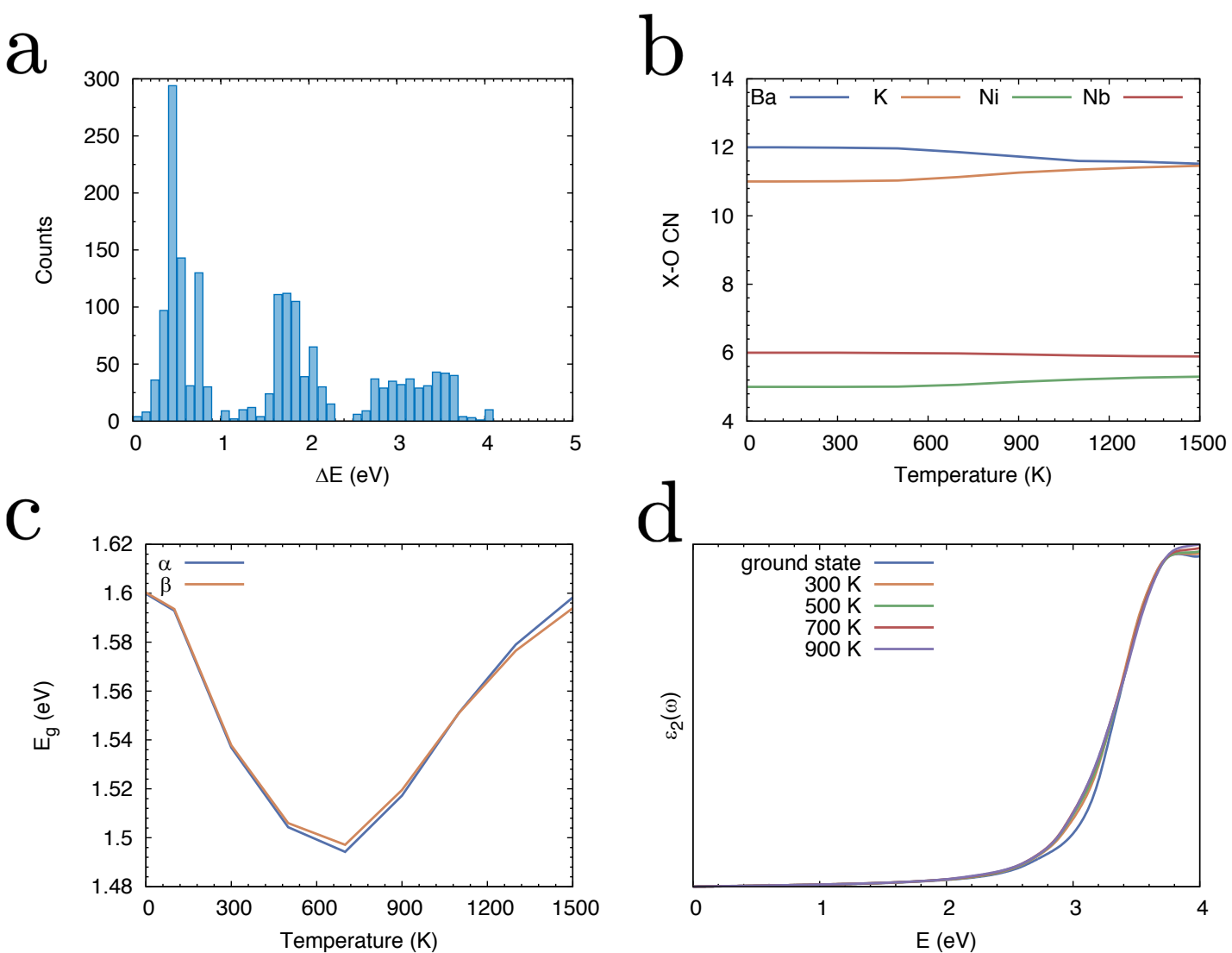


Figure 1. **a**) Histogram with relative energies, ΔE , for all the structures explored for the $0.5\{\text{KNbO}_3\}-0.5\{\text{Ba}(\text{Ni}_{0.5}\text{Nb}_{0.5})\text{O}_{3-\delta}\}$ system; **b**) Coordination number, CN, for each cation, X, at different temperatures; **c**) PBE+U band gap, E_g , for the majority, α , and minority, β , spin at different temperatures; **d**) Simulated spectra using the imaginary part of the dielectric function at different temperatures.